Coordination Compounds

Study notes · Chemistry

Contents

- Coordination Compounds 1
- Practice questions 2

★ Coordination Compounds

► 1.1 Introduction and nomenclature

- **Coordination entity:** A central metal atom or ion surrounded by ligands, such as $[Fe(CN)_6]^{3-}$.
- **Ligand:** An ion or molecule that donates an electron pair to the central metal ion.
  - Examples include H_2O, NH_3 and Cl^-.
- **Coordination number:** The number of donor atoms attached to the central metal.

> **याद रखें**
>
> A ligand may be monodentate, bidentate or polydentate. दाता परमाणुओं की संख्या पर ध्यान दें।

► 1.2 Crystal field splitting

The five d orbitals split into two sets in an octahedral field. The energy difference is represented by Δo.

$$\mu_s = \sqrt{n(n+2)}$$

> **MAGNETIC MOMENT**
>
> **Question:** Find the spin-only magnetic moment for three unpaired electrons.
> **Solution:** μs = √(3(3+2)) = √15 ≈ 3.87 BM.

► 1.3 Comparing geometries

Property	Octahedral	Tetrahedral
Coordination number	6	4
Splitting	Δo	Δt ≈ 4Δo/9
Examples	$[Fe(CN)_6]^{3-}$	$[NiCl_4]^{2-}$

★ Practice questions

Q1. Which ligand is bidentate?

(A) H_2O (B) ethylenediamine

(C) NH_3 (D) Cl^-

Q2. Calculate the oxidation state of Fe in $[Fe(CN)_6]^{3-}$.